

Advanced neuro ophthalmology

Parts 1 and 2 | Demonstrate the deficit before naming its cause

Afferent and motor functions localise different parts of the pathway.

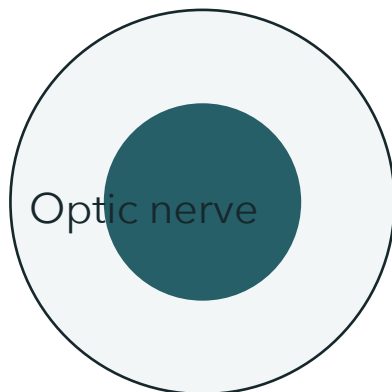
Finding	Implication
Afferent testing	Acuity, colour, fields and pupil asymmetry
Disc and OCT	Swelling, axonal change and structural mimics
Motility and sensation	Cranial-nerve and orbital compartment clues
Targeted imaging	Answer the unresolved localisation question

Reduced colour vision, a compatible field defect and an RAPD support unilateral optic neuropathy even when central acuity remains good. Added ophthalmoplegia and trigeminal sensory loss suggest neighbouring orbital-apex involvement. A slowly progressive isolated neuropathy raises sheath compression; contrast MRI should address that compartment. Disc appearance alone cannot separate true neuropathy from buried structural mimics.

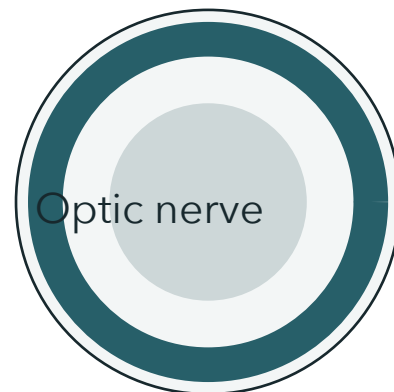
Inflammation is a family of phenotypes

A positive antibody does not replace a compatible syndrome.

Intraneural pattern



Sheath pattern



Enhancement localisation narrows the differential; it does not uniquely establish the cause.

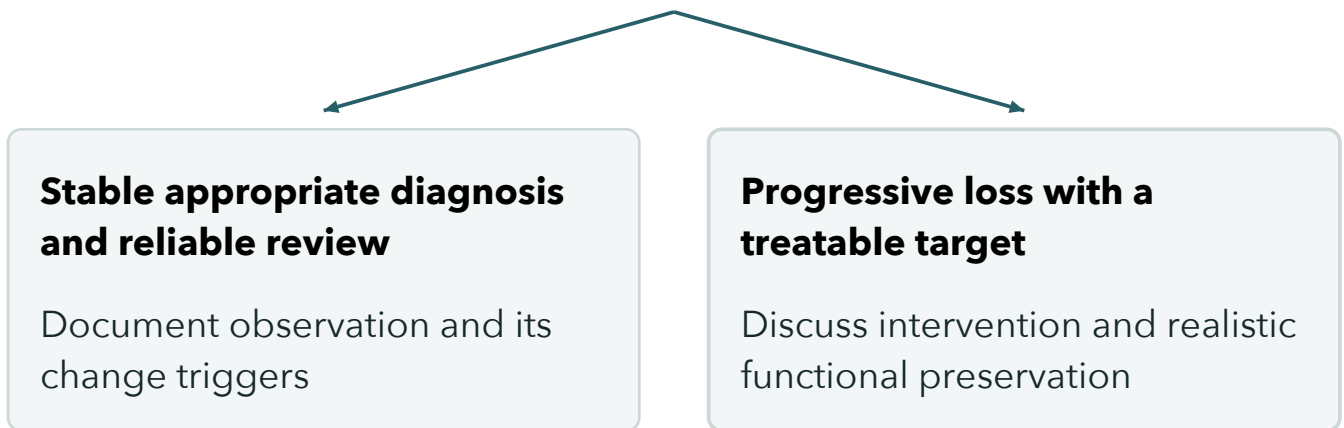
Original compartment sketch: intraneural and sheath enhancement are distinct observations.

Typical MS-associated optic neuritis is often unilateral and relatively short segment. MOG-associated disease often involves long anterior segments, marked disc swelling, bilateral disease or perineural enhancement; AQP4 disease favours posterior pathways and chiasmal involvement. Interpret MOG cell-based assays within the clinical syndrome, particularly low-positive results. Sheath enhancement also requires evaluation for non-MOG inflammatory or infiltrative disease.

A longitudinal decision needs comparable measurements

Stable disease and progressive useful-vision loss can justify different plans.

Is function demonstrably changing?



For an optic nerve sheath meningioma with stable useful function, comparable acuity, colour and field measurements support observation. Documented progressive loss may justify fractionated radiotherapy to preserve remaining function; established axonal loss limits recovery. LHON requires a different discussion: carrier status and heteroplasmy do not precisely predict penetrance or an individual relative's clinical course.

Diplopia: distinguish geometry from pathway injury

Restriction, paresis and central imbalance need different explanations.

Finding	Implication
Central torsional-vertical pattern	Consider skew and associated neurological signs
Crossed fourth-nerve pathway	Nuclear laterality differs from peripheral nerve
Myopic globe-muscle displacement	Assess mechanical vector change
Fatigable variable weakness	Consider neuromuscular transmission

A skew deviation may combine vertical misalignment with conjugate torsion and other brainstem signs; the hypertropic eye is often intorted rather than following a simple fourth-nerve pattern. Trochlear fibres cross, so a nuclear lesion affects the opposite superior oblique. Fatigability supports neuromuscular transmission, while high-myopic globe displacement can alter muscle paths without a primary nerve palsy.

Evidence and immediate systemic risk

A plausible treatment can have an unfavourable benefit-risk balance.

Finding	Implication
Pituitary apoplexy with instability	Empirical cortisol replacement may be urgent
Disabling ocular myasthenia	Discuss indicated immunotherapy and monitoring
Indirect TON with head injury	Routine megadose steroid benefit is unproven; harm matters
Relapsing MOG disease	Distinguish cohort support from randomised proof

Pituitary apoplexy with haemodynamic instability requires empirical hydrocortisone without delaying treatment for endocrine confirmation. Functionally limiting ocular myasthenia may warrant immunotherapy when pyridostigmine is insufficient, with assessment for generalisation. Maintenance IVIG in relapsing MOG-associated disease has supportive cohort evidence; this is distinct from randomised proof and from immediate treatment of an unstable endocrine emergency.